CS 5610 Project 2: Design Document

PhotoFind

Project Description

Images and Search is Rishabh's (my) half of PhotoFind, a web application for organizing, finding, and planning photos. PhotoFind has two independent features, each with its own MongoDB collection:

1. **Images and Search** (Rishabh Kumar): add images, tag them, and search for them.
2. **Shot Lists** (Milena Perez-Gerus): plan the shots for a trip and track your progress.

People collect photos from many places (their phone, their camera, links saved from the web) and then can't find the one they want without scrolling through everything. Images and Search keeps them in one place and makes them findable by what's in them. Users add an image by uploading a file or pasting a URL, and give it a few short tags and an optional description. Uploaded photos are resized in the browser before saving, so large photos stay small in the database. The gallery can be filtered by tag, any image can be edited or deleted, and the search page ranks images by how well their tags and description match what the user types, showing a % match for each result.

The images collection holds over 1,000 records. The two features use separate collections, so neither depends on the other.

User Personas

Persona 1: Amy the Weekend Traveler

- **Background**: A nurse who comes home from every trip with hundreds of photos.
- **Goals**: Keep her trip photos in one place and find specific shots later without endless scrolling.
- **Pain Points**: Her photos pile up on her phone, and finding one shot from a trip months ago means scrolling through thousands.
- **Behavior**: Sorts through her photos on her days off after a trip and keeps only the ones worth saving.

Persona 2: Leo the Graduate Student

- **Background**: An environmental science student whose field photos are spread across his phone, a camera, and links shared by his lab.
- **Goals**: Describe each field photo so he can find the right one when writing a report.
- **Pain Points**: File names like IMG_4821.jpg say nothing about the site or the conditions in the photo.
- **Behavior**: Adds notes right after each field day, while he still remembers the details.

Persona 3: Sam the Hobby Designer

- **Background**: A graphic designer who saves reference images from all over the web.
- **Goals**: Tag references by color and style so he can pull up "that blue poster" when he needs it.
- **Pain Points**: Bookmarks and screenshot folders pile up with no way to search them by what they show.
- **Behavior**: Saves references in quick bursts while browsing, then reorganizes them as his collection grows.

User Stories

User Story 1: Amy saves her trip photos

As Amy, I want to upload photos from my computer, so that my trip pictures are saved in one place.

After her Boston weekend, Amy opens PhotoFind and clicks "+ Add image." She chooses a photo of the Charles River from her computer. Her phone took it at 3000 × 2000 pixels, but the preview shows it was resized to 1024 × 683, about 92 KB. She clicks Save and lands on the image's page with an "Image added" message.

User Story 2: Leo describes his field photos

As Leo, I want to add tags and a short description to each image, so that I can find it later by what's in it.

After a field day, Leo uploads a photo of a riverbank site. He types the tags "river, erosion, site 3" and watches them appear as separate badges as he types. In the description he writes "East bank after heavy rain, visible soil loss." Months later the photo is still easy to find, even though its file name was meaningless.

User Story 3: Sam browses by tag

As Sam, I want to filter the image grid by a tag, so that I can see all related images at once.

Sam is starting a new project with a blue color scheme. In the gallery he clicks the "blue (14)" tag button, and the grid shows only his 14 blue references. He copies the page link, which now ends in ?tag=blue, and sends it to a colleague, who sees the same filtered view.

User Story 4: Sam fixes his tags

As Sam, I want to change an image's tags or description, so that I can fix mistakes or reorganize as my collection grows.

Sam notices that a poster he tagged "blu" never shows up under "blue." He opens the image, clicks Edit, corrects the tag and adds "minimal," then clicks Save. The page updates right away with a "Changes saved" message, and the poster now appears under the blue filter.

User Story 5: Amy cleans up her collection

As Amy, I want to delete images I no longer want, so that my collection stays clean.

Amy finds a blurry duplicate of a sunset photo. She opens it and clicks Delete, and a confirmation popup makes sure she doesn't remove the wrong one by accident. She confirms and is taken back to the gallery, which shows "Image deleted."

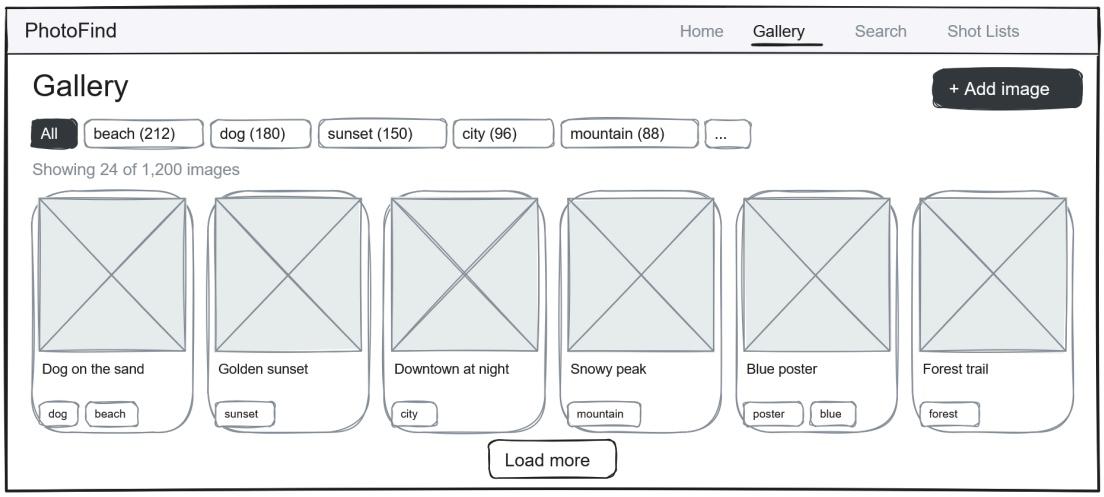
User Story 6: Sam finds "that blue poster"

As Sam, I want to type what I remember into a search bar and see matching images, so that I can find that one desired image quickly.

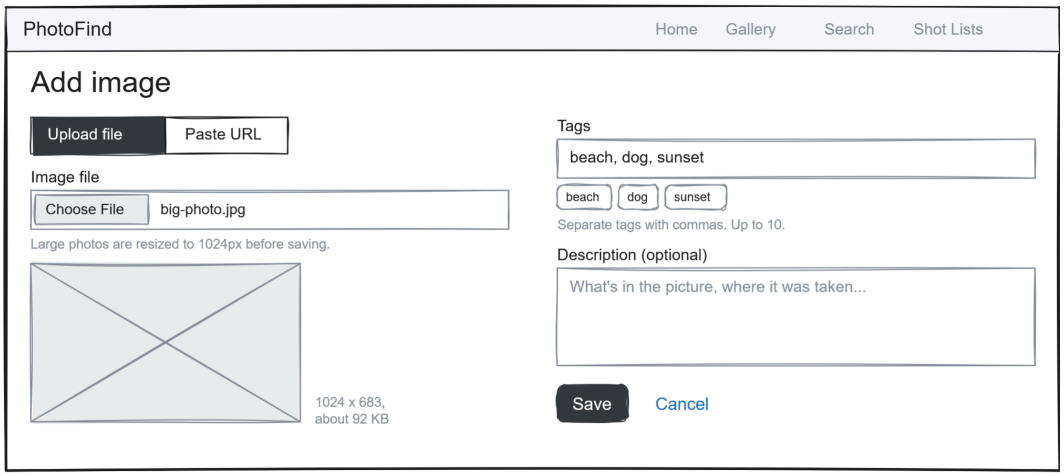
Sam remembers a reference but not when he saved it. He opens Search and types "blue poster minimal." The poster appears first with a green "100% match" badge, and the matching words are highlighted in its description. Below it are other images that matched only some of his words, with lower percentages. He finds it in seconds instead of scrolling.

Wireframes

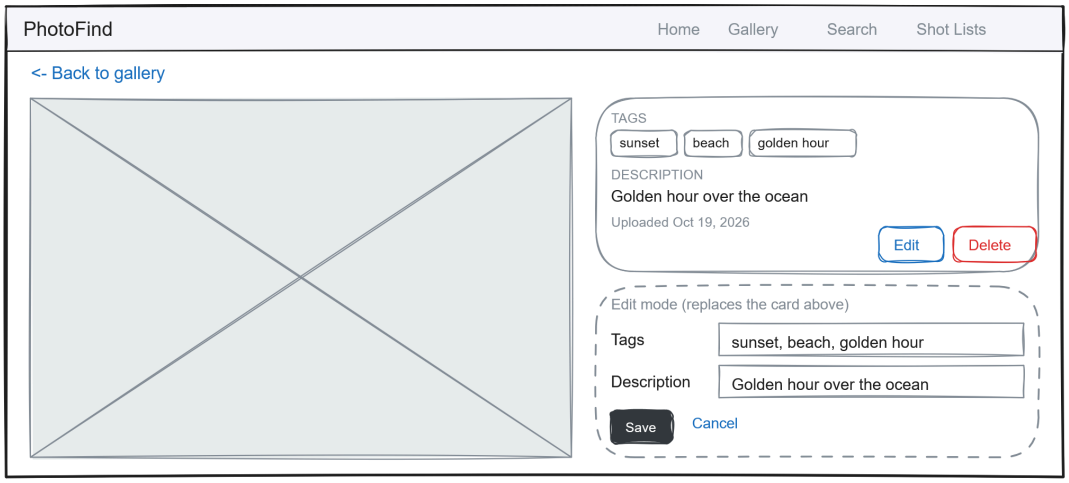
1. Gallery



2. Add Photo



3. Image Details



4. Search

